Polar Movement Relative to Australia

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Summary

During Upper Proterozoic times the mean geomagnetic pole seems to have been situated near Australia. By the Cambrian it appears to have moved to what is nowadays the region of South Africa passing southwards into the Southern Ocean by Silurian and early Devonian times. In the Upper Carboniferous and Permian the pole lay in the Tasman Sea and Australia was once again in a high geomagnetic latitude. Later the pole moved slowly southwards, reaching the present geographic pole towards the end of the Tertiary.

1. Introduction

The mean position of the geomagnetic poles during any geological epoch may be calculated from estimates of the mean direction of magnetization of rock formations laid down at that time (Creer, Irving & Runcorn 1957). The method of obtaining these estimates has been discussed previously (Fisher 1953, Watson & Irving 1957, Runcorn 1957). These pole positions are given relative to the area in which the formation is situated. If rock formations from successive geological periods are studied in the same general region the path of polar wandering relative to this region is obtained. From palaeomagnetic surveys of some rock formations in Australia the movements of the geomagnetic pole relative to Australia during and since the Upper Proterozoic has been obtained in broad outline. The data are very far from complete but sufficient is known for certain general points to be made. The path of movement is known in some detail from Carboniferous to Recent times and with less certainty for periods prior to this. The stability of the magnetization of several of the rock formations studied has been demonstrated (Irving 1956, 1957a) by tests of the type described by Graham (1949).

Samples have been collected from rock formations distributed over a wide area of the Commonwealth (Figure 1). Most of the samples have been collected by the authors with invaluable geological guidance from Dr W. R. Browne of Edgecliff, Sydney, Professor S. W. Carey and Mr M. R. Banks of the University of Tasmania, Dr O. P. Singleton of the University of Melbourne, and Dr A. A. Opik and Mr L. C. Noakes of the Bureau of Mineral Resources. In addition, many samples have been sent to us by officers of the Bureau of Mineral Resources by arrangement with Dr N. H. Fisher, and the Geological Survey of Western Australia by arrangement with Mr H. A. Ellis. The following officers of these organisations have collected specimens for us: Messrs M. A. Randall, R. Stuart and N. Mackay of the Bureau of Mineral Resources, and Mr L. de la Hunty of the Geological Survey of Western Australia. The specimens of Brisbane tuff

were sent to us by Mr I. Macleod of the University of Queensland. We wish to express our gratitude to these people for their very great help, and also to Professor J. C. Jaeger, Head of this Department, for many helpful suggestions and criticisms.

A list of the formations studied in Australia with their mean directions of magnetisation and attendant pole positions is given in the table. Each result is numbered in order of age from the oldest to the youngest, the numbering in the

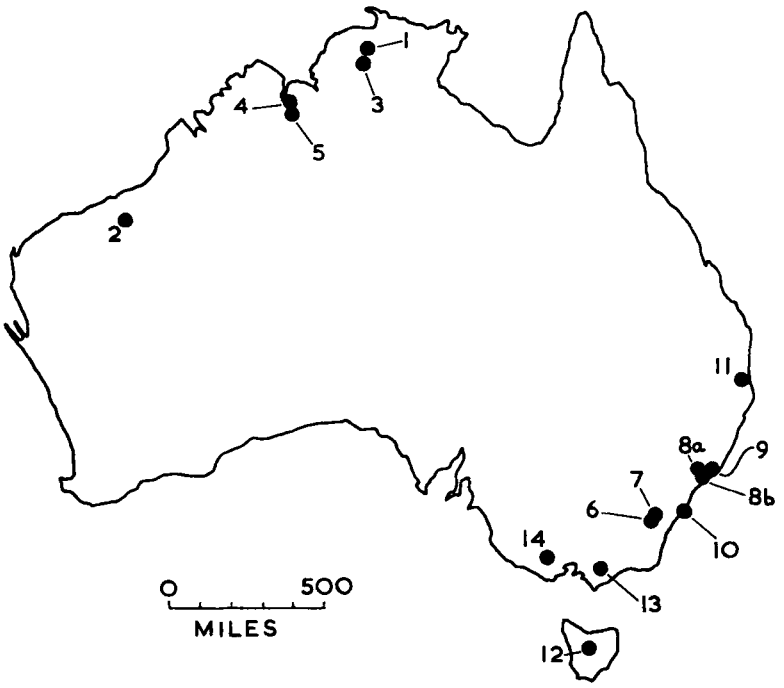


FIG. 1.—*Locations of the rock formations sampled for palaeomagnetic work.* The formations are numbered as in the table.

table and the figures being consistent. Correction is always made for geological dip so that the directions are given relative to the bedding (or, in the case of igneous rocks, the bedding of adjacent sediments) which is assumed horizontal at the time of deposition. Reversals of magnetization occur in many of the formations, for instance in the Antrim Plateau Basalts, and in these cases the two alternative directions are given. For those rock formations in which reversals have not so far been detected, the north-seeking directions are given. Only the southern hemisphere poles are quoted in the table and there are, of course, antipoles to these in the northern hemisphere.

Stereographic projections are used throughout. On these the mean pole position is indicated by a dot and is surrounded by an error area within which the pole lay at $P = 0{\cdot}95$. The area of error is circular when the sampling region was near the ancient pole, for example in the Permian volcanics, but is oval when the sampling region lay near the ancient equator. For a given sample size the former is eight times the area of the latter so that accurate pole determinations from rock formations deposited in high latitude require a comparatively large number of specimens.

Table of rock formations (all angles are in degrees).

No.	Formation	Geological age	Mean direction of magnetisation			Sampling Area		Pole position			
			D	I	α	Lat.	Long.	Lat.	Long.	$d\psi$	$d\chi$
14	Newer volcanics of Victoria	Pliocene, Pleistocene and Recent	3·4 183·4	−59·8 +59·8	4·8	38·0S	143·5E	86·3S	102·1E	5·5	7·2
13	Older volcanics of Victoria	Lower Tertiary, probably Eocene	17·0 197·0	−72·9 +72·9	6·8	38·0S	145·5E	66·8S	122·7E	10·5	12·1
12	Dolerite sills of Tasmania	Mesozoic, probably Jurassic	325·0	−85·0	3·5	42S	147E	50S	157E	6·8	6·8
11	Brisbane tuff	Triassic, probably lower Triassic	35 215	−83 +83	—	27·5S	153·5E	39S	143E	—	—
10	Volcanics of Illawarra coast	Permian, Upper Marine Series	67	+81	11	35S	151E	27S	169E	21	21
9	Volcanics of Hunter Valley	Permian, Lower Marine Series	110	+80	—	33S	151E	38S	174E	—	—
8b	Kuttung red varvoid sediments	Upper Carboniferous	270 90	−84 +84	6	33S	151E	32S	165E	12	12
8a	Kuttung Lavas	Upper Carboniferous	5	−85	8	33S	151E	43S	150E	16	16
7	Ainslie volcanics	Devonian, probably lower Devonian	17	−30	12	35S	149E	66S	12E	7	13
6	Mugga phorphyry	Upper Silurian	26	−30	22	35S	149E	60S	23E	14	24
5	Elder Mountain sandstone	Middle Cambrian	231	−15	10	16S	126E	34S	8E	5	10
4	Antrim plateau basalts	Lower Cambrian	53 233	− 2 + 2	12	16S	126E	36S	26E	6	12
3	Buldiva quartzite	Top of Upper Proterozoic	243	+38	12	14S	132E	30S	59E	8	14
2	Nullagine lavas	Upper Proterozoic	143	+64	8	21S	120E	51S	162E	10	13
1	Edith River volcanics	Lower part of Upper Proterozoic	90	+48	18	13S	132E	6S	166W	15	24

D is the declination reckoned east of geographic north; I is the inclination considered positive when below the bedding plane; α is the semi-angle of the cone of confidence around the mean direction at $P = 0{\cdot}95$ (Fisher 1953, Watson & Irving 1957); $d\psi$ and $d\chi$ are the semi-axes of the oval error area round the pole position at the same probability, $d\psi$ is the error in co-latitude (Creer, Irving & Runcorn 1957). In formation No. 12 reversals have been recorded, but they are of restricted occurrence.

2. Polar movement since the Carboniferous

The pole during the Upper Carboniferous (Nos. 8a and 8b) determined from the sediments and lavas of the Kuttung Series of the Hunter River valley lay in what is nowadays the Tasman Sea (Figure 2). In Permian times (Nos. 9 and 10) it lay a little to the north and east. It is interesting that this result confirms an early measurement by Mercanton (1926) who recorded an almost vertical magnetization in a single specimen from these lavas. By Mesozoic times the pole passes

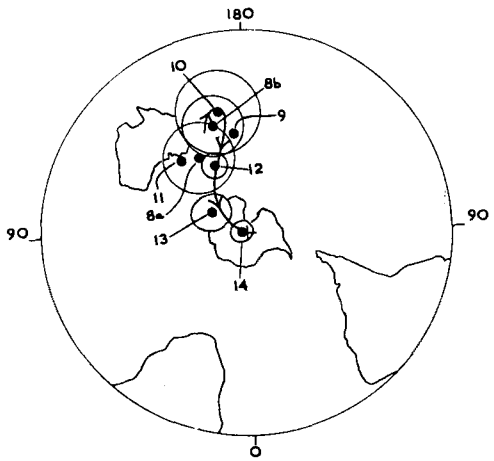


FIG. 2.—*Polar movement since the Carboniferous.* The poles are numbered as in the Table. The southern hemisphere poles are plotted, the antipoles being given in Figure 5. The approximate pole path is indicated in thick line (Polar projection).

into the Southern Ocean (Nos. 11 and 12) reaching the coast of Antarctica in the Lower Tertiary (No. 13). Since then it has coincided with the present South geographic pole (No. 14). The Northern hemisphere polar path for this time passes from North-west Africa in a broad sweep through the North Atlantic to the North geographic pole (Figure 5).

There is a suggestion in Figure 2 that the Permian pole is in advance of the Carboniferous one so that the direction of polar movement may have reversed during this time. However, the pole errors are large and verification by future measurements is necessary, but it is of interest that in the determinations from North America and Europe the Carboniferous pole positions also appear to be somewhat in advance of the Permian poles (see Figure 5, and also a review by Irving (1957b)).

3. Polar movement prior to the Carboniferous

For earlier times the pole path is sketched only in preliminary fashion (Figure 3). Three rock formations within the Upper Proterozoic of Australia (Adelaidian System) have been studied. The oldest are the Edith River Volcanics, a formation within the Catherine River Group, which belongs to the lower part of the Upper Proterozoic (Noakes 1956); the pole lay in the mid-Pacific (No. 1). The Nullagine lavas are part of a sequence which contains evidence of glaciation and for this reason are correlated tentatively with the Sturtian stage of the Adelaidian System (David 1950, p. 72), that is, about the middle of the Upper Proterozoic; the pole

determined from these lavas lay near New Zealand (No. 2). The relative stratigraphic positions of the Edith River Volcanics and these Nullagine lavas is uncertain and it is assumed here that the latter are somewhat younger as seems most likely from the glaciation evidence. The Buldiva quartzite, which contains fossils of primitive life, is the youngest Proterozoic formation studied. It is placed at the very top of the Upper Proterozoic and is sometimes referred to as sub-Cambrian (Noakes 1956). The pole (No. 3) lay near Mauritius in the South Indian Ocean.

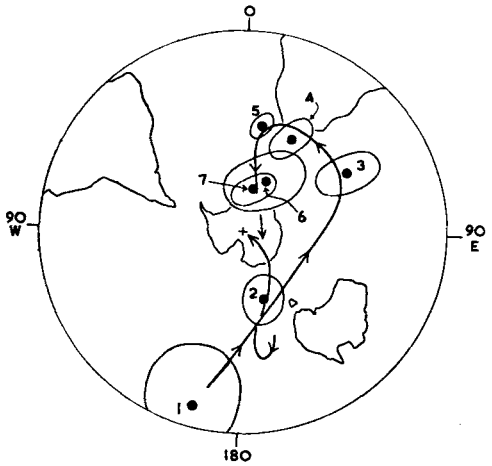


FIG. 3.—*Polar movement during and since the Upper Proterozoic.* The poles are numbered as in the Table. The pole path sketched is necessarily conjectural. The path of polar movement since Carboniferous times is added in thinner line for comparison. The possible linkage of these two through Antarctica is indicated (Polar projection).

The Upper Proterozoic appears to have been a period of very considerable polar wandering as the palaeomagnetic data from the Northern hemisphere also indicates (Du Bois 1955, Creer & others 1957).

The pole during Lower and Middle Cambrian times lay near South Africa (Nos. 4 and 5). During the late Silurian and early Devonian it was situated in a similar longitude but well south in the Southern Ocean.

Because of the occurrence of reversals of magnetization the palaeomagnetic method, although it determines the position of the pole, does not give its polarity. Clearly, unless an approximately continuous pole sequence is obtained, as for instance in Figure 2, an unambiguous path cannot be traced. In the Australian results there are very great discontinuities in the Upper Proterozoic and especially in the Upper Palaeozoic between the Lower Devonian and the Upper Carboniferous, and ambiguities of this type consequently arise. Clearly, the curve given in Figure 3 is only one of many possible polar paths (it is in fact the simplest possible path) and further sampling will be necessary before these uncertainties are settled.

4. Palaeoclimatic aspects

There are theoretical reasons connected with the origin of the geomagnetic field for supposing that the average geocentric dipole will always be directed along the Earth's axis of rotation (Runcorn 1954, Creer & others 1957). Palaeomagnetic evidence from the Upper Cenozoic in both hemispheres has shown that this has been the case during the last 20 million years (Hospers 1955, Irving & Green 1957). It is therefore possible that the pole positions given in this paper may also be the

positions of the geographic poles relative to Australia. If this is so the palaeoclimatic conditions in Australia should be broadly consistent with the geomagnetic latitude deduced from the palaeomagnetic data.

The variations in geomagnetic latitude for South-eastern Australia are given in Figure 4. During the Upper Proterozoic, and again in the late Palaeozoic and Mesozoic, Australia was in a high geomagnetic latitude and it is at these times that the geological evidence indicates a cold climate. The tillites and varved sediments

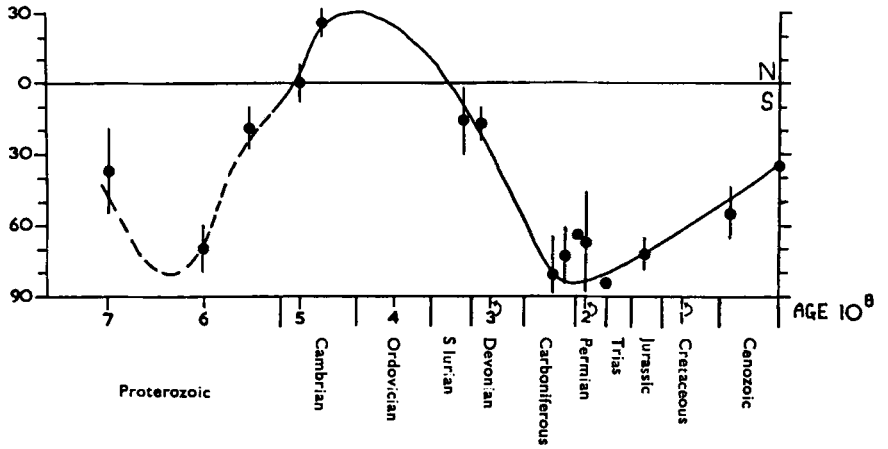


FIG. 4.—*The variations in geomagnetic latitude in South-eastern Australia.* These curves are calculated for a central point in South-eastern Australia (35S, 145E). The errors are obtained from the errors in the determinations of the positions of the ancient poles and represent the limits within which the latitude lay at $P = 0{\cdot}95$. The curve is derived from Figures 2 and 3.

of the Adelaidian System and the outwash gravels of the Nullagine rocks provide evidence of extensive glaciation in the earlier period (David 1950, p. 79). Similar deposits in the Kuttung Series and its equivalents elsewhere in Australia indicate a glaciation in the Upper Carboniferous (David 1950, p. 326). Evidence of glaciation is again found in the Permian (David 1950, pp. 341, 397), and the deposits of the Triassic, Jurassic and Cretaceous are generally regarded as having been laid down in cold to temperate conditions (David 1950, pp. 440, 476, 515). The fossil trees during these periods have pronounced growth rings suggesting the well-marked seasonal changes characteristic of higher latitudes. The great saurian reptiles are absent and the invertebrate marine faunas have a cold water aspect (see, for instance, Whitehouse 1926). The indications of warm climate which are widespread in other parts of the world at these times, for example in the Mediterranean region, are absent, and this would seem to imply that Australia was in high or intermediate latitudes just as the palaeomagnetic results do. It may also be noted that from the Proterozoic to Devonian the geomagnetic latitudes are low, and during these times no glacial formations are known in Australia, whereas red beds, dolomites, thick limestones with algal, Archaeocyathid and coral reefs are common.

Because of exceptional meteorological conditions in the past the palaeoclimatic evidence may not always give a close indication of latitude. Nevertheless the broad correspondence between this evidence and the palaeomagnetic latitude, in particular the coincidence of high geomagnetic latitude with glaciation, suggest that on average the geomagnetic and geographic poles have always been coupled together,

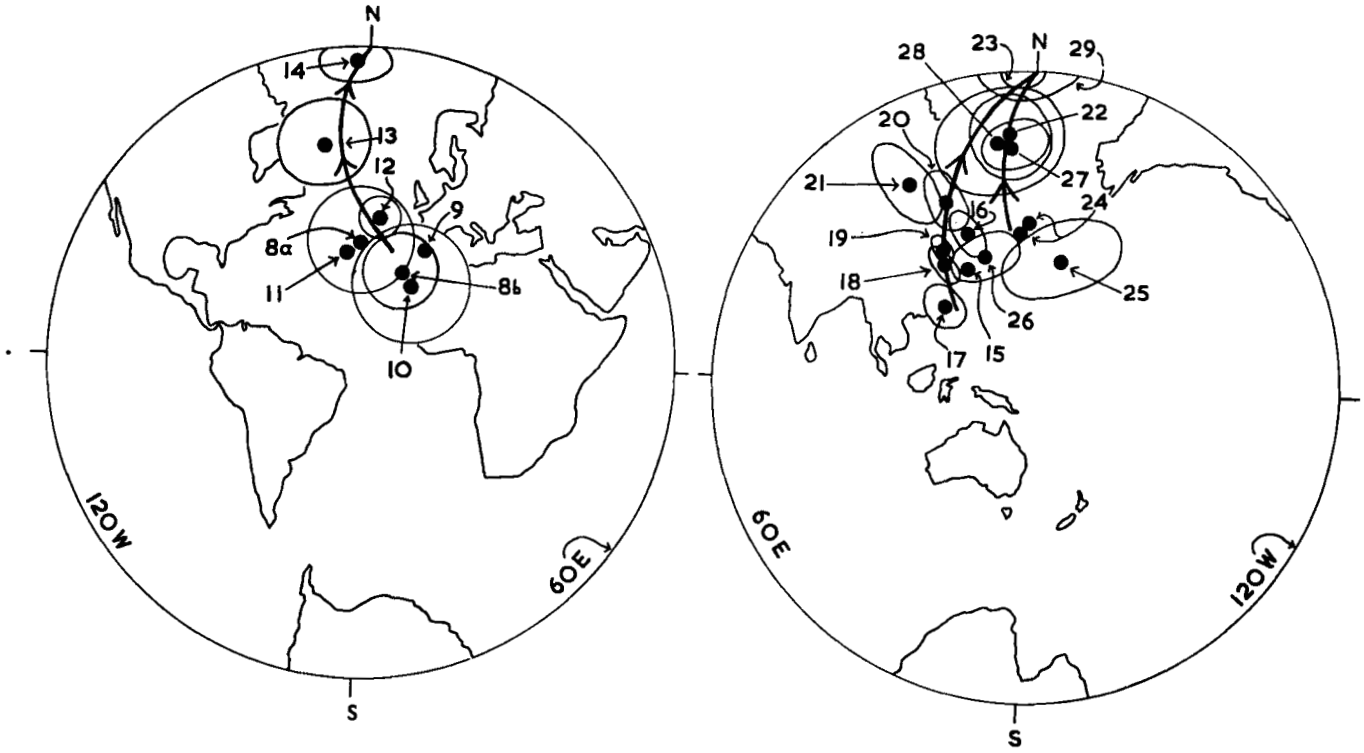


FIG. 5.—*Pole positions obtained from rock formations of Carboniferous and later ages in North America, Europe and Australia (equatorial projection).* The large discrepancy between equivalent results from Australia and those from northern continents is illustrated here. The individual poles for Australia are numbered as in the Table. The North American poles are numbered as follows:

15. Mississipian (Howell & Martinez 1957)
16. Pennsylvanian (Runcorn 1956)
17. Permian (Runcorn 1956)
18. Permian (Doell 1955)
19. Permian (Graham 1955)
20. Triassic (Runcorn 1956)
21. Triassic (Du Bois *et. al.* 1957).
22. Cretaceous (Runcorn 1956)
23. Upper Tertiary (Campbell & Runcorn 1956)

The European poles have these numbers:

24. Lower Carboniferous (Belshe 1955)
25. Permian (Creer & others 1957)
26. Triassic (Clegg, Almond & Stubbs 1954)
27. Eocene (Hospers 1955)
28. Oligocene (Roche 1950)
29. Miocehe (Hospers 1955).

and that so far as Australia is concerned the great Proterozoic and late Palaeozoic glaciations may be reasonably explained by movement of the geographic poles relative to Australia. The ambiguity mentioned at the end of the previous section does not alter this argument. It should be noticed from Figure 2 that this explanation for glaciations in Australia would not provide for the late Palaeozoic glaciations of South America, South and Central Africa and India, assuming that the present continental positions have always been maintained.

5. Comparison with palaeomagnetic results from elsewhere

Except in later Cenozoic times the poles determined from the Australian data do not coincide with equivalent results from Europe, North America, Africa and India. The contrast is greatest in the Upper Palaeozoic and early Mesozoic where the discrepancies between pole determinations from rock formations of similar age is between 50° and 90° of arc, and are far greater than any experimental error (Figure 5). It would seem very difficult to explain these discrepancies without entertaining the possibility that in the past Australia has moved as a whole relative to the Northern continents.

The polar paths for Carboniferous and later times obtained from North America, Europe and Australia have an approximately similar form. They are roughly meridional in direction, begin in similar latitudes, and converge on the present pole in later Tertiary times. Their different longitudinal positions could be explained by supposing that relative movements have occurred between these continents since the Carboniferous, movements which possibly extended into the Tertiary (Irving & Green, 1957). However, the form of the polar path from Australia prior to the Carboniferous is so different from that obtained from the Northern continents (Creer & others 1957) that the supposition of post-Carboniferous continental drift *only* is inadequate and it is necessary to suppose that relative movements also occurred prior to the Carboniferous. *If* the discrepancies between equivalent pole results from different continents are to be explained by continental drift then there seems to be no reason to regard it as a unique process occurring only at one stage of geological history; in fact there are suggestions that it was also operative in earlier epochs.

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References

Belshe, J. C., quoted by Runcorn, S. K. (1955). *Adv. in Phys.*, **4**, 244.
Campbell, C. D. & Runcorn, S. K., 1956. *Journ. Geophys. Res.*, **61**, 449.
Clegg, J. A., Almond, M. & Stubbs, P. H. S., 1954. *Phil. Mag.*, **45**, 583.
Creer, K. M., Irving, E. & Runcorn, S. K., 1957. *Phil. Trans. Roy. Soc.* A, **250**, 144.
David, Sir T. W. E., 1950. *The Geology of the Commonwealth of Australia*, (edit. by Browne, W. R.), **1**. Arnold, London.
Doell, R. R., 1955. *Nature* (London), **176**, 1167.
Du Bois, P. M., 1955. *Nature* (London), **176**, 506.

Du Bois, P. M., Irving, E., Opdyke, N. D., Runcorn, S. K. & Banks, M. R., 1957. *Nature* (London), **180**, 1186.
Fisher, Sir R. A., 1953. *Proc. Roy. Soc.* A, **217**, 295.
Graham, J. W., 1949. *Journ. Geophys. Res.*, **54**, 131.
Graham, J. W., 1955. *Journ. Geophys. Res.*, **60**, 329.
Hospers, J., 1955. *Journ. Geol.*, **63**, 59.
Howell, L. G. & Martinez, J. D., 1957. *Geophysics*, **22**, 384.
Irving, E., 1956. *Pap. and Proc. Roy. Soc. Tasmania*, **90**, 157.
Irving, E., 1957a. *Nature* (London), **180**, 280.
Irving, E., 1957b. *Adv. in Phys.*, **6**, 194.
Irving, E. & Green, R., 1957. *Nature* (London), **179**, 1064.
Mercanton, P. L., 1926. *Terr. Mag.*, **31**, 187.
Noakes, L. C., 1956. XXth Int. Geol. Congress, Mexico. Symposium on the Cambrian and its lower boundary, **2**, 213.
Roche, A., 1950. *C. R. Acad. Sci.*, Paris, **230**, 113.
Runcorn, S. K., 1954. *Trans. Amer. Geophys. Un.*, **35**, 49.
Runcorn, S. K., 1956. *Bull. Geol. Soc. Amer.*, **67**, 301.
Runcorn, S. K., 1957. *Adv. in Phys.*, **6**, 169.
Watson, G. S. & Irving, E., 1957. *Mon. Not. R. Astr. Soc. Geophys. Suppl.*, **7**, 289.
Whitehouse, F. W., 1926. *Mem. Qld. Mus.*, **5**, 195.